MSc Data Science

DS601: Mathematical Foundations for Data Science

Date: 10/08/2026

Question 1: Independence of Events on a Coloured Tetrahedron

A regular tetrahedron (4-sided die) has one solid red face, one solid blue face, one solid green face, and one face containing all three colours. Let R, B, and G be the events that the face landing face-down contains Red, Blue, and Green, respectively.

1. Concept Used: Pairwise Independence and Mutual Independence

Two events A and B are independent when $P(A \cap B) = P(A)P(B)$. Three events R, B, and G are pairwise independent if this condition holds for every pair. They are mutually independent only if the pairwise conditions hold and, in addition, $P(R \cap B \cap G) = P(R)P(B)P(G)$.

2. Approach

Because the tetrahedron is regular, each face has probability 1/4. First calculate P(R), P(B), and P(G), then check every pair. Finally, test the three-way intersection.

3. Step-by-Step Solution

Step 1: Each colour appears on two faces: its solid-colour face and the mixed face. Therefore,

$$P(R) = P(B) = P(G) = 2/4 = 1/2.$$

Step 2: Red and Blue occur together only on the mixed face, so

$$P(R \cap B) = 1/4 = (1/2)(1/2) = P(R)P(B).$$

Similarly,

$$P(R \cap G) = 1/4 = P(R)P(G), \quad \text{and} \quad P(B \cap G) = 1/4 = P(B)P(G).$$

Therefore, R, B, and G are pairwise independent.

Step 3: All three colours occur together only on the mixed face. Hence,

$$P(R \cap B \cap G) = 1/4.$$

But

$$P(R)P(B)P(G) = (1/2)(1/2)(1/2) = 1/8.$$

Since $1/4 \neq 1/8$, the three events are not mutually independent.

Final answer: R, B, and G are pairwise independent, but they are not mutually independent.

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Question 2: Finding Probabilities of Two Independent Events

Let A and B be independent events. The probability that at least one occurs is 0.88, and the probability that exactly one occurs is 0.56. Find P(A) and P(B).

1. Concept Used: Union Formula, Independence, and Exactly-One Probability

For two events, P(A B) = P(A) + P(B) - P(A ∩ B). Independence gives P(A ∩ B) = P(A)P(B). For exactly one event to occur, the overlap must be excluded completely, so P(exactly one) = P(A) + P(B) - 2P(A ∩ B).

2. Approach

Let P(A) = x and P(B) = y. Use the two given probabilities to form two equations. Subtract the equations to obtain xy, then obtain x + y and solve the resulting quadratic equation.

3. Step-by-Step Solution

Step 1: Since A and B are independent, P(A ∩ B) = xy. The “at least one” condition gives

$$x + y - xy = 0.88. \quad (1)$$

Step 2: “Exactly one” means A occurs without B or B occurs without A. Therefore,

$$x + y - 2xy = 0.56. \quad (2)$$

Step 3: Subtract Equation (2) from Equation (1):

$$xy = 0.88 - 0.56 = 0.32.$$

Step 4: Substitute xy = 0.32 into Equation (1):

$$x + y - 0.32 = 0.88, \text{ so } x + y = 1.20.$$

Step 5: Numbers x and y with sum 1.20 and product 0.32 are roots of

$$t^2 - 1.20t + 0.32 = 0.$$

Factorising,

$$(t - 0.80)(t - 0.40) = 0.$$

Hence the two probabilities are 0.80 and 0.40. The information does not distinguish which value belongs to A and which belongs to B.

Final answer: {P(A), P(B)} = {0.80, 0.40}; equivalently, P(A) = 0.80 and P(B) = 0.40, or vice versa.

MSc Data Science

DS601: Mathematical Foundations for Data Science

Date: 10/08/2026

Question 3: Conditional Probability Without Replacement

A jar contains 6 red marbles and 4 blue marbles. Two marbles are drawn sequentially without replacement. Find the probability that the second marble is red, given that the first marble is blue.

1. Concept Used: Conditional Probability and Sampling Without Replacement

The phrase “given that the first marble is blue” means we restrict attention to the situation after a blue marble has already been removed. Because the drawing is without replacement, the composition of the jar changes after the first draw.

2. Approach

Condition on the first draw being blue. Remove one blue marble from the original contents, then calculate the fraction of the remaining marbles that are red.

3. Step-by-Step Solution

Step 1: Initially, the jar contains 6 red + 4 blue = 10 marbles.

Step 2: We are told that the first marble drawn is blue. After removing that blue marble, the jar contains:

6 red marbles and 3 blue marbles, for a total of 9 marbles.

Step 3: Of the 9 remaining marbles, 6 are red. Therefore,

$$P(\text{second red} \mid \text{first blue}) = 6/9 = 2/3 \approx 0.6667.$$

Thus, once we know that the first draw was blue, the probability that the second draw is red is about 66.67%.

Final answer: P(second red | first blue) = 2/3 ≈ 66.67%.

Class-activity interpretation (66 students): If 66 students are arranged into 33 pairs and each pair performs one fresh two-draw experiment, then the probability that a pair gets Blue first and Red second is

$$P(B_1 \cap R_2) = P(B_1)P(R_2 \mid B_1) = (4/10)(6/9) = 4/15 \approx 26.67\%.$$

Therefore, among 33 independent pair-trials, the expected number of Blue-then-Red outcomes is $33 \times 4/15 = 8.8$, i.e., about 9 pairs.

MSc Data Science

DS601: Mathematical Foundations for Data Science

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Question 4: Two-Leg Digital Transmission and Bayes' Theorem

A source transmits 1 with probability 0.6 and 0 with probability 0.4. The bit passes through two independent transmission legs. On each leg, the bit flips with probability 0.1. If the final receiver observes a 1, find the probability that the source originally transmitted a 1.

1. Concept Used: Conditional Probability, Law of Total Probability, and Bayes' Theorem

To reverse the direction of conditioning, Bayes' theorem is used: $P(S=1 \mid R=1) = P(R=1 \mid S=1)P(S=1) / P(R=1)$. The denominator $P(R=1)$ is found by the law of total probability, because the source must have sent either 1 or 0.

2. Approach

First calculate the probability that the receiver sees 1 given that the source sent 1. Then calculate the same probability given that the source sent 0. Use these two cases to find $P(R=1)$, and finally apply Bayes' theorem.

3. Step-by-Step Solution

Step 1: Find $P(R=1 \mid S=1)$. Starting from 1, the receiver ends at 1 if there are zero flips or two flips:

$$P(R=1 \mid S=1) = (0.9)(0.9) + (0.1)(0.1) = 0.81 + 0.01 = 0.82.$$

Step 2: Find $P(R=1 \mid S=0)$. Starting from 0, the receiver ends at 1 if exactly one of the two legs flips:

$$P(R=1 \mid S=0) = (0.1)(0.9) + (0.9)(0.1) = 0.09 + 0.09 = 0.18.$$

Step 3: Use the law of total probability to find the overall probability that the receiver observes 1:

$$P(R=1) = P(R=1 \mid S=1)P(S=1) + P(R=1 \mid S=0)P(S=0)$$

$$= (0.82)(0.60) + (0.18)(0.40) = 0.492 + 0.072 = 0.564.$$

Step 4: Apply Bayes' theorem:

$$P(S=1 \mid R=1) = [(0.82)(0.60)] / 0.564 = 0.492/0.564 \approx 0.87234.$$

Therefore, after observing a final 1, the probability that the source originally transmitted 1 is approximately 87.23%.

Final answer: $P(S=1 \mid R=1) \approx 0.8723 \approx 87.23\%$.

MSc Data Science
DS601: Mathematical Foundations for Data Science
Date: 10/08/2026

Question 5: Two Factories and Exactly One Defective Product

A product is produced by F1, F2, or F3. The factories produce 20%, 30%, and 50% of the products, with defective probabilities 1%, 2%, and 4%, respectively. Two products are selected independently. Find the probability that exactly one of the two products is defective.

1. Concept Used

Law of Total Probability and Independence

2. Approach

First find the probability that one product is defective. Then find the probability that it is non-defective. Exactly one defective can occur in two cases: defective then non-defective, or non-defective then defective.

3. Step-by-Step Solution

Step 1: Find the probability of a defective product

$P(D) = P(F1)P(D|F1) + P(F2)P(D|F2) + P(F3)P(D|F3)$
$= (0.20)(0.01) + (0.30)(0.02) + (0.50)(0.04)$
$= 0.002 + 0.006 + 0.020 =$ **0.028**.

Step 2: Find the probability of a non-defective product

$P(N) = 1 - P(D) = 1 - 0.028 =$ **0.972**.

Step 3: First defective, second non-defective

$P(D,N) = P(D)P(N) = (0.028)(0.972) =$ **0.027216**.

Step 4: First non-defective, second defective

$P(N,D) = P(N)P(D) = (0.972)(0.028) =$ **0.027216**.

Step 5: Add the two possible cases

$P(\text{exactly one defective}) = P(D,N) + P(N,D)$
$= 0.027216 + 0.027216 =$ **0.054432**.
Therefore, the probability is **5.4432%**.

Final answer: P(exactly one defective) = 0.054432 = 5.4432%.

MSc Data Science
DS601: Mathematical Foundations for Data Science
Date: 10/08/2026

Question 6: Student Late on a Randomly Chosen Trip

A student travels to college using Bus (50%), Auto (30%), or Bike (20%). Without rain, the late probabilities are 0.10, 0.20, and 0.05. With rain, they are 0.30, 0.40, and 0.15. The probability of rain is 30%, and transport choice is independent of rain. Find the probability that the student is late.

1. Concept Used

Law of Total Probability and Independence

2. Approach

First calculate the probability of being late when there is no rain and when it rains. Then combine the two cases using the probabilities of rain and no rain.

3. Step-by-Step Solution

Step 1: Probability of no rain

P(No Rain) = 1 - 0.30 = **0.70**.

Step 2: Probability of being late when there is no rain

P(L|No Rain) = (0.50)(0.10) + (0.30)(0.20) + (0.20)(0.05)
= 0.05 + 0.06 + 0.01 = **0.12**.

Step 3: Probability of being late when it rains

P(L|Rain) = (0.50)(0.30) + (0.30)(0.40) + (0.20)(0.15)
= 0.15 + 0.12 + 0.03 = **0.30**.

Step 4: Combine rain and no rain

P(L) = P(No Rain)P(L|No Rain) + P(Rain)P(L|Rain)
= (0.70)(0.12) + (0.30)(0.30)
= 0.084 + 0.090 = **0.174**.
Therefore, the probability is **17.4%**.

Final answer: P(L) = 0.174 = 17.4%.

MSc Data Science
DS601: Mathematical Foundations for Data Science
Date: 10/08/2026

Question 7: Two Balls Drawn Without Replacement

A box contains 6 Red, 5 Blue, and 4 Green balls. Two balls are drawn successively without replacement.

1. Concept Used

Conditional Probability and Sampling Without Replacement

2. Approach

For each part, condition on the first draw when necessary. Since the balls are drawn without replacement, the total number and color counts change after the first draw.

3. Step-by-Step Solution

Step 1: Probability that both balls are Red

Total balls = 6 + 5 + 4 = 15.
P(R1) = 6/15.
After one red is drawn, 5 red balls remain out of 14:
P(R2|R1) = 5/14.
Therefore, P(both Red) = (6/15)(5/14) = **1/7**.

Step 2: Given the first ball is Red, probability that the second is Blue

After removing one Red, 14 balls remain, including 5 Blue balls.
P(B2|R1) = 5/14 = **5/14**.

Step 3: Given that a uniformly selected one of the two drawn balls is observed to be Blue, probability that it was the first ball

Case 1: First Blue, second not Blue:
P(B1, not B2) = (5/15)(10/14) = 5/21. Selecting the first ball has probability 1/2, so the contribution is 5/42.

Case 2: Both Blue:
P(B1,B2) = (5/15)(4/14) = 2/21. Selecting the first ball has probability 1/2, so the contribution is 1/21.

Thus P(first selected Blue) = 5/42 + 1/21 = 1/6.

The probability that the selected ball is Blue is
(5/21)(1/2) + (5/21)(1/2) + 2/21 = 1/3.

Therefore,
P(first ball | selected ball is Blue) = (1/6)/(1/3) = **1/2**.

Step 4: Probability that the two balls are of different colors

Red-Blue:
P(RB or BR) = (6/15)(5/14) + (5/15)(6/14) = 2/7.

Red-Green:
P(RG or GR) = (6/15)(4/14) + (4/15)(6/14) = 8/35.

Blue-Green:
P(BG or GB) = (5/15)(4/14) + (4/15)(5/14) = 4/21.

Therefore,
P(different colors) = 2/7 + 8/35 + 4/21 = **74/105**.

Final answer: 1. 1/7 2. 5/14 3. 1/2 4. 74/105.

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Question 8: Two Fair Dice — Win, Bonus, and Immediate Loss

A player rolls two fair dice. The player wins if the sum is 7 or 11, receives a bonus if both dice show the same number, and loses immediately if the sum is 3 or less. Find: (a) the probability of winning, (b) the probability of receiving a bonus or losing immediately, and (c) whether the three probabilities can simply be added to get 1.

1. Concept Used

Basic Probability, Union of Events, and Overlap

2. Approach

There are 36 equally likely outcomes. For each part, identify the relevant outcomes and account for any overlap between events.

3. Step-by-Step Solution

(a) Probability that the player wins

Sum 7 outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): 6 outcomes.
Sum 11 outcomes: (5,6), (6,5): 2 outcomes.
Therefore, P(Win) = (6 + 2)/36 = 8/36 = **2/9**.

(b) Probability that the player receives a bonus or loses immediately

Bonus means both dice are equal: (1,1), (2,2), ..., (6,6), so P(Bonus) = 6/36.

Immediate loss means sum is 2 or 3. Sum 2 has (1,1): 1 outcome. Sum 3 has (1,2), (2,1): 2 outcomes. Hence P(Lose) = 3/36.

The outcome (1,1) belongs to both events, so it must be subtracted once:
P(Bonus or Lose) = 6/36 + 3/36 - 1/36 = 8/36 = **2/9**.

(c) Can the three probabilities simply be added to get 1?

No. The events are not mutually exclusive because (1,1) is both a bonus and an immediate loss. Therefore, simply adding the three probabilities counts (1,1) twice. Also, some outcomes are neither a win, nor a bonus, nor an immediate loss; for example, (2,4) has sum 6 and the dice are different.

Thus the probabilities cannot simply be added to get 1.

Final answer: (a) 2/9 (b) 2/9 (c) No.